

PROJECT BLUEPRINT: THE NIGER DELTA ENVIRONMENTAL BIG DATA AUDIT

Exhibition Destination: Rivers State University (RSU) Faculty of Science Conference

1. THE BIG DATA TOOLKIT ARCHITECTURE

Your project splits your 6-month training portfolio into 4 clean layers:

- **Ingestion & Compute Layer: Python** running locally or via a script. It uses **NumPy** and **Scikit-Learn** to generate 1,000 parallel streams from mock environmental sensors without using your laptop's memory.
 - **Storage & Database Layer: AWS RDS (Relational Database Service)** hosting a **PostgreSQL** database engine. This provides the cloud hardware, scalable RAM, and disk indexing to process millions of incoming records securely over the internet.
 - **Analytics & Predictive ML Layer: Google Colab** (Cloud Notebook) connected to AWS via **SQLAlchemy**. You will use **Scikit-Learn** classical machine learning models (like Linear Regression) to generate short-term predictive data points.
 - **Visualization Layer: Streamlit** built entirely in Python. It bypasses corporate account restrictions, hooks directly to AWS, executes code dynamically, and is hosted for free on **Streamlit Community Cloud**. **Microsoft Excel** links via **Power Query** to pull quick baseline aggregates.
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2. THE 5-PHASE EXECUTION PLAN

Phase 1: Cloud Provisioning (Setting up AWS RDS)

- **Objective:** Provision your remote data warehouse engine.
- **Action Steps:** Log into your AWS Console, navigate to the RDS Dashboard, and spin up a free-tier PostgreSQL database instance. Adjust the inbound security rules to allow traffic from anywhere (0.0.0.0/0) so your laptop and your hosted app can communicate with it via its unique remote AWS Endpoint URL.

Phase 2: Database Architecture & Partitioning (SQL)

- **Objective:** Structure tables to handle heavy data velocity without query lag.
- **Action Steps:** Connect to your cloud server using pgAdmin 4. Write an optimized SQL layout script to generate a master stream table. Configure Table Partitioning by List to slice the main dataset into independent physical tables grouped by Rivers State Local Government Areas (e.g., Eleme, Bonny, Port-Harcourt).

Phase 3: Generative Synthesizer & Stream (Python)

- **Objective:** Generate true big data velocity using statistically accurate synthetic data.

- **Action Steps:** Load the real Port Harcourt Kaggle dataset into a Python script. Train a Scikit-Learn Gaussian Mixture Model (GMM) to learn its covariance, variance, and feature correlations. Write a loop that samples from this model to create 30,000 completely unique rows every 3 seconds, dynamically applies a live timestamp, and bulk-writes the micro-batch into your AWS RDS database.

Phase 4: Analytical Modeling & Predictive ML (Python)

- **Objective:** Discover patterns and predict future environmental shifts.
- **Action Steps:** Open a Google Colab cloud notebook, connect to your AWS RDS database instance, and run advanced aggregations using Common Table Expressions (CTEs) and Window Functions. Train a Scikit-Learn Linear Regression model to calculate an Air Quality Index (AQI) forecast based on the incoming generative data streams. Write these results back to a predictive table in your cloud DB.

Phase 5: Streamlit Web UI Deployment (Python App)

- **Objective:** Deliver an elegant, live dashboard for the conference judges.
- **Action Steps:** Write a concise Streamlit app script using pure Python. Use the database connector to query your AWS database dynamically based on user selections. Incorporate responsive input widgets (sliders and dropdowns) alongside Plotly map overlays. Commit your code to GitHub and link it to Streamlit Community Cloud to generate a live, shareable public web application link.

3. THE CONFERENCE EXHIBITION PITCH

When the panel of Faculty of Science judges walks up to your booth, deliver this exact narrative structure:

1. **The Limitation Pitch:** *"Traditional tools like local Excel files crash at 1 million rows and cannot handle continuous sensor feeds. This architecture overcomes hardware limitations through cloud computing."*
 2. **The Generative Machine Learning Aspect:** *"Instead of completely fake data, I trained a Scikit-Learn Gaussian Mixture Model on historical Port Harcourt air quality records. It learns the mathematical DNA of real soot pollution to generate an infinite, unique stream of 30,000 rows every 3 seconds—providing true big data velocity."*
 3. **The Infrastructure Proof:** *"A regular database would freeze at this speed. I used Table Partitioning inside AWS RDS PostgreSQL to shard data by LGA. This allows my Streamlit web dashboard to query millions of rows in the cloud and refresh live metrics right in front of you instantly."*
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